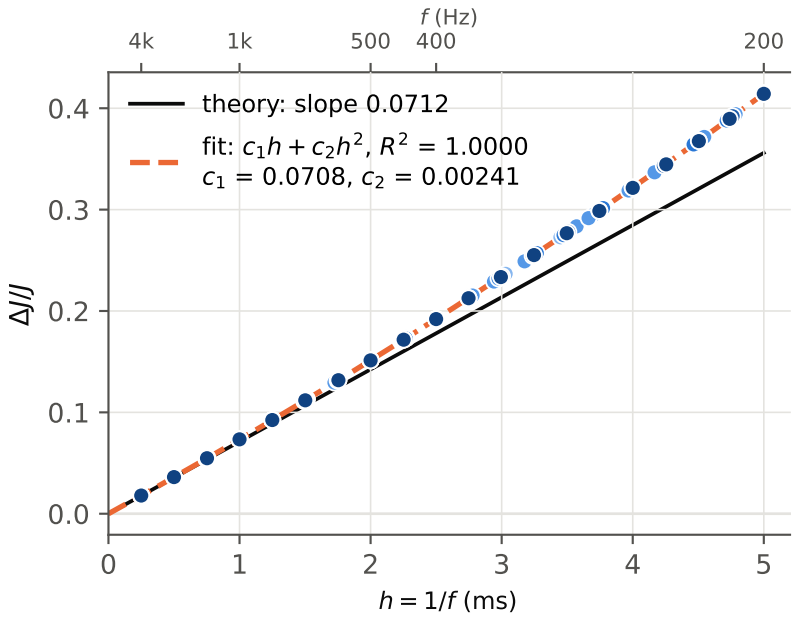
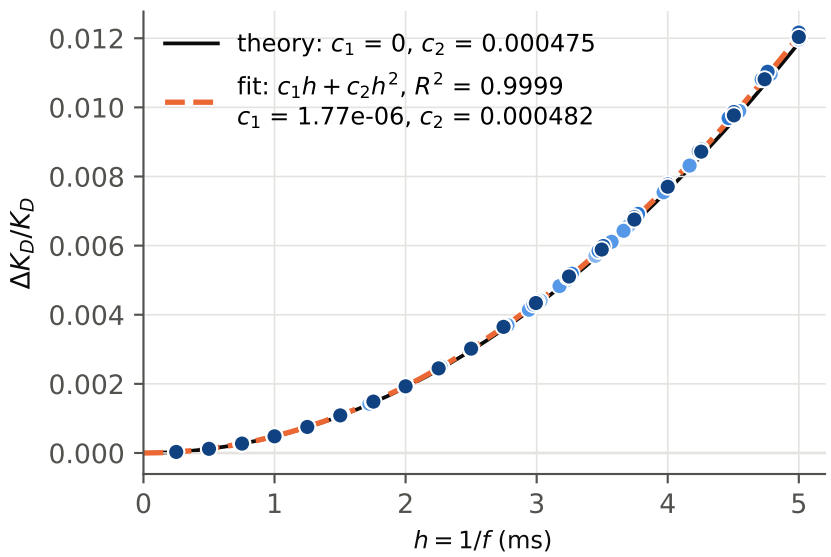
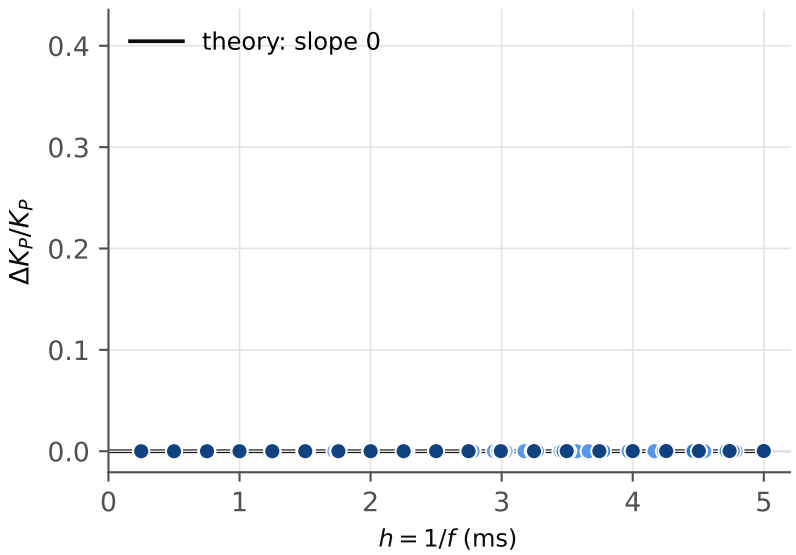
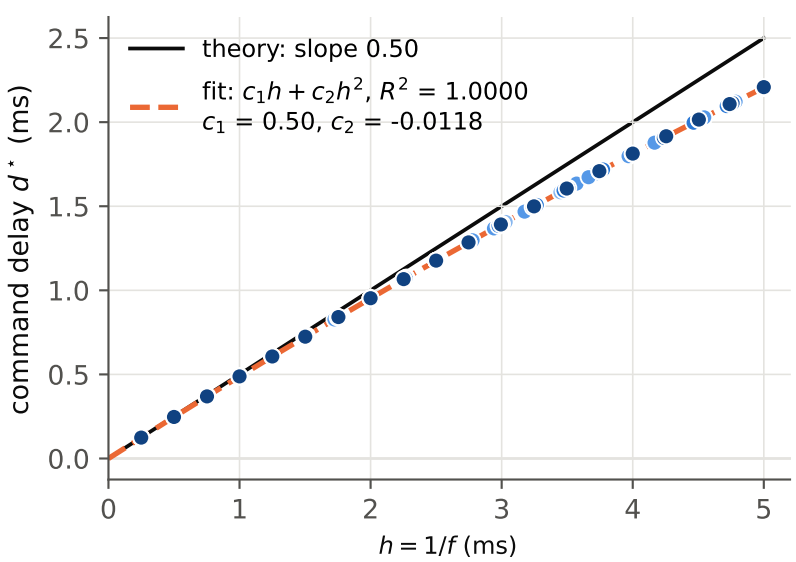
$f_{ctrl} \approx$ 10 Hz · $f_{ctrl} \approx$ 20 Hz · $f_{ctrl} \approx$ 50 Hz · $f_{ctrl} \approx$ 100 Hz · $f_{ctrl} \approx$ 200 Hz

## A: Euler + explicit PD
fitted: $J, K_P, K_D, d$

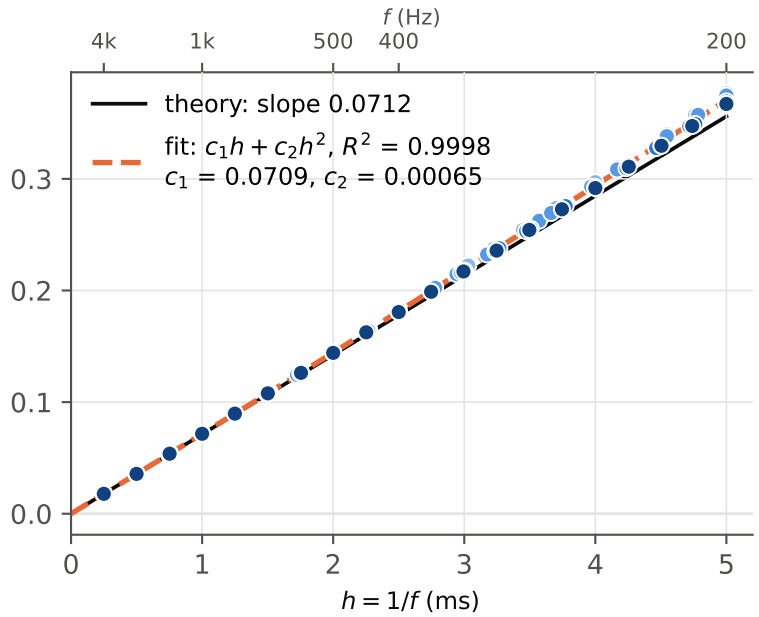


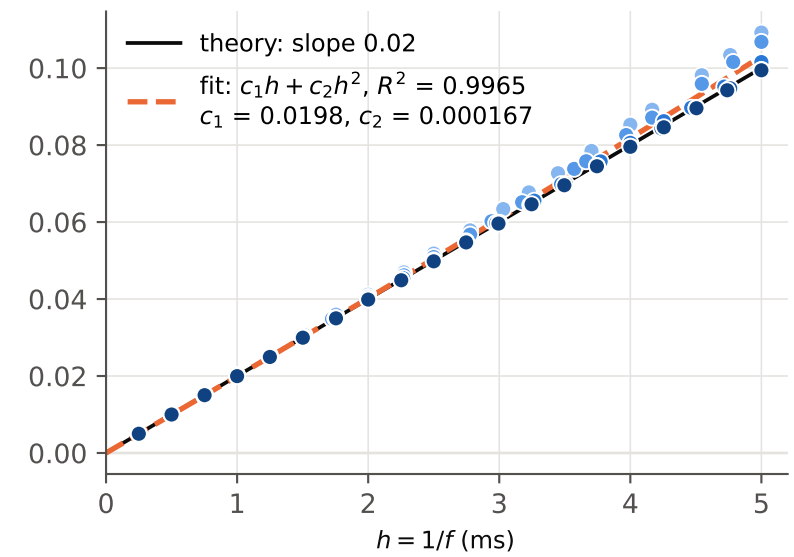



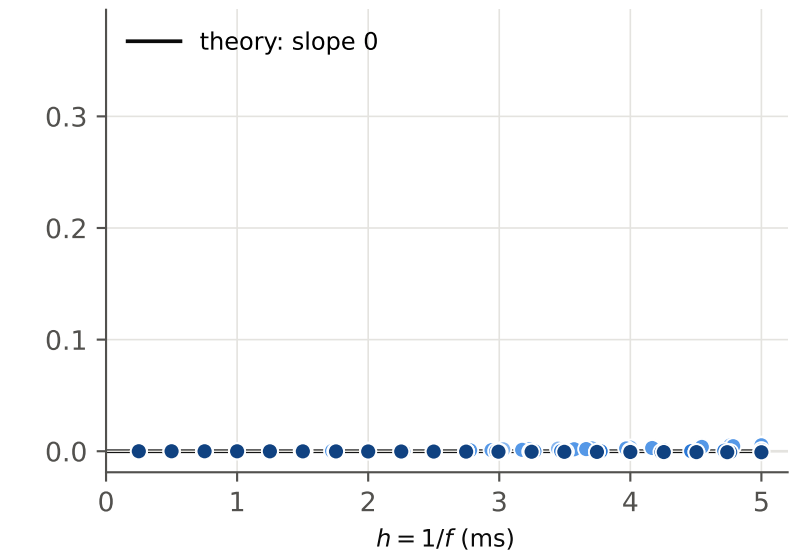



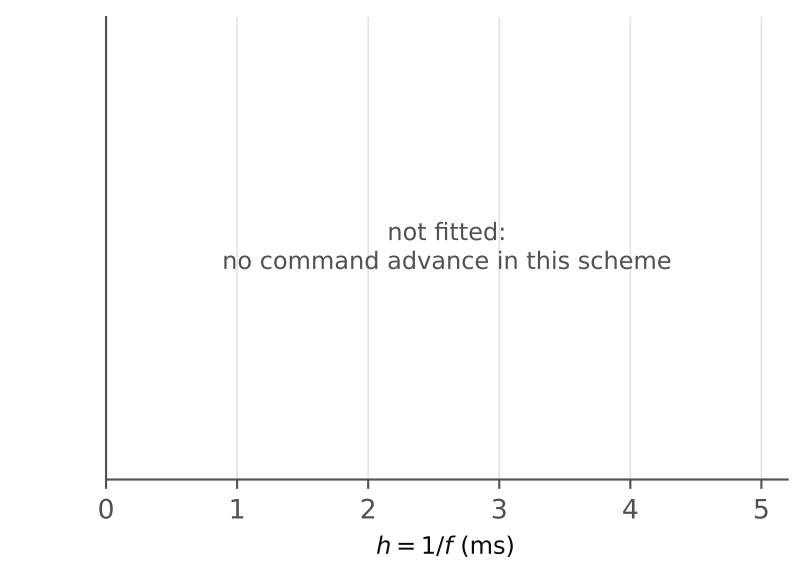




## B: RK4 + explicit PD (torque held)
fitted: $J, K_P, K_D$

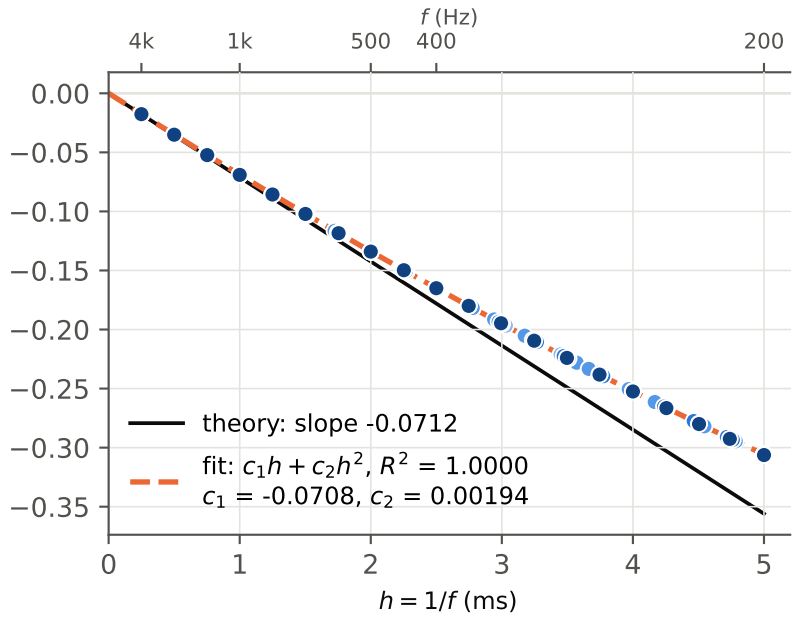


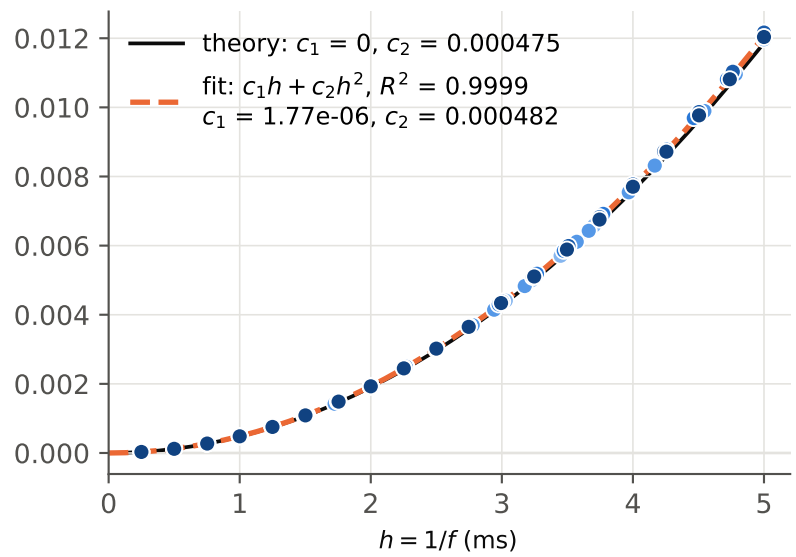



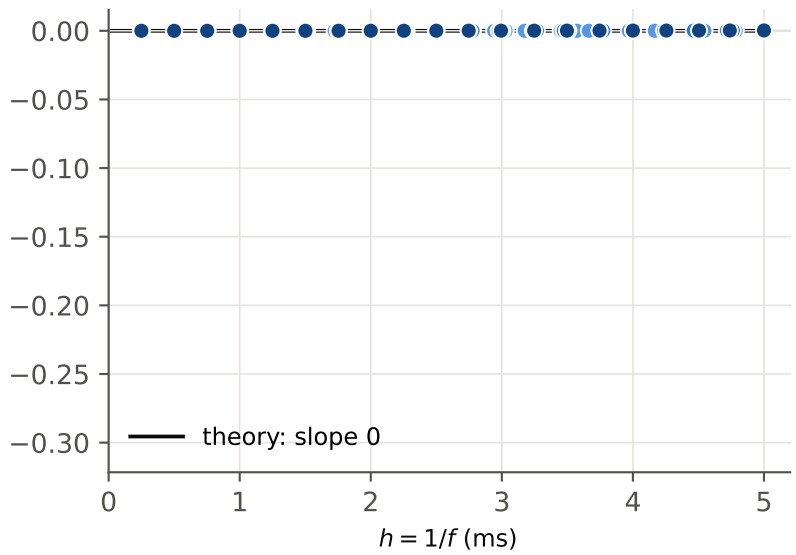



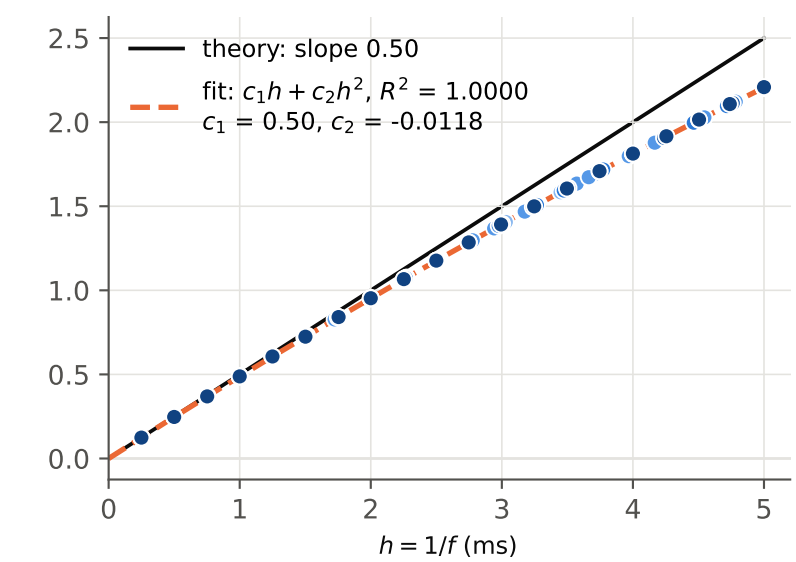




## C: implicitfast + native PD
fitted: $J, K_P, K_D, d$




Target $h$ = 0.25, 0.5, ..., 5 ms ($f$ = 200-4000 Hz), realized with integer $f_{ctrl}$ (Hz) and integer decimation, within 2.3% of the target.
Theory (solid), first order except the $K_D$ row of A and C, which carries the derived $h^2$ counterterm $K_PK_D/12J$.
Fit (dashed): $c_1h + c_2h^2$ through the origin, so $c_1$ is the measured first-order slope.